



Gyanmanjari Innovative Unniversity

Gyanmanjari Degree Engineering College

Department of Computer Science and Engineering

Semester-3

Database Management System: Organizing, Querying, and Managing Data – BET1CE13305

Name of Faculty: Dr. Mansi C. Shanishwara

Hrs. Week: 6 hours

Micro Session Planner

Session no.	Module	Description	Time	Teacher aid	Planned date	Actual date	Remarks
Module-1 : Introduction to Database Management Concept and SQL Basics							
1	1	Topics: Introduction to DBMS and its Terminologies					
		Subject Introduction and Evaluation Methodology	20 Min	Explanation, Smartboard			
		Oracle Installation	40 Min	Smartboard, Laptop			
		Introduction of DBMS	10 Min	PPT, Smartboard.			
		Introduction, Data, Information	15 Min	PPT, Smartboard.			
		Data Items, Fields, Records, Files,	20 Min	PPT, Smartboard.			
		Data Dictionary, Metadata, Database, Database Systems	15 Min	PPT, Smartboard.			
		Homework: 1. Complete Oracle Installation if pending. 2. Prepare data dictionary of Student Table. 3. Prepare data dictionary of Employee Table.					
2	1	Topics: DBMS Purpose and Applications, Oracle Database and RDBMS Architecture					
		Revision	10 Min	Discussion			
		Database management system: Introduction, Purpose, Applications	15 Min	PPT, Smartboard.			
		Introduction to Oracle Database, RDBMS Architecture	10 Min	PPT, Smartboard.			
		Explanation of SEE-1 and CCE of Module 1: Student Instructions for Performing the Activity and Submitting the Required Work	15 Min	Explanation			
		Practical 1: Implement SQL queries for Creating and describing tables (CREATE, DESC)	15 Min	SQL Plus			
		Practical 9: Create student tables with a few columns representing common data types (integer, text, date etc.) using DDL commands.	15 Min	SQL Plus			
		Practical Assignment-1 Problem Solving and Practice.(1 - 5 Questions)	40 Min	SQL Plus			
3	1	Topics: Open source and Commercial DBMS, DataTypes in Oracle, DDL Commands					
		Revision	15 Min	Discussion			
		Open source and Commercial DBMS (MYSQL, ORACLE, DB2, SQL server)	10 Min	PPT, Smartboard.			
		Basics of SQL: Data Types in Oracle	10 Min	PPT, Smartboard.			
		About SQL Commands and its types	15 Min	PPT, Smartboard.			
		SQL Categories: Data Definition Language (DDL) Commands: CREATE, ALTER, DROP	20 Min	PPT, Smartboard., SQL Plus			
		Practical 2: Implement SQL queries for Modifying tables using ALTER	10 Min	SQL Plus			
		Practical 3: Implement SQL queries for Deleting tables using DROP	5 Min	SQL Plus			
		Practical Assignment-1 Problem Solving and Practice. (6-14 Questions)					
		Homework:					
		1. Create a table named EMPLOYEE with fields: employee_id (number), employee_name (varchar), dob (date), and email (varchar).					
		2. Create a table named DEPARTMENT with fields: department_id, department_name, location, and budget.					
		3. Alter the EMPLOYEE table to add a column phone_number (varchar).					
		4. Rename the column email in the EMPLOYEE table to employee_email.					
		5. Drop the DEPARTMENT table.					



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4	1	Topics: DML Commands					
		Revision	10 Min	Discussion			
		Data Manipulation Language (DML) Commands: INSERT	15 Min	PPT, Smartboard., SQL Plus			
		Data Manipulation Language (DML) Commands: UPDATE	10 Min	PPT, Smartboard., SQL Plus			
		Data Manipulation Language (DML) Commands: DELETE	10 Min	PPT, Smartboard., SQL Plus			
		Practical 4: Implement SQL queries for create table and insert sample data in tables.	15 Min	SQL Plus			
		Practical 8: Create Two tables' Students and Employees with 5 fields and stored actual data into the table using create and insert command.	15 Min	SQL Plus			
		Practical 5: Implement SQL queries for Inserting and updating records using INSERT, UPDATE, DELETE	10 Min	SQL Plus			
		Practical Assignment-1 Problem Solving and Practice. (15-20 Questions)	35 Min	SQL Plus			
		Homework:					
		1. Insert 5 records into the LIBRARY table with sample data.					
		2. Insert 3 records into the DEPARTMENT table					
5	1	Topic : DQL Commands, Clauses : Where, Order by					
		Revision	10 Min	Discussion			
		Data Query Language (DQL) Command: SELECT with WHERE Clause	15 Min	PPT, Smartboard., SQL Plus			
		DQL Command: SELECT with ORDER BY	20 Min	PPT, Smartboard., SQL Plus			
		Aggregate Functions Examples	20 Min	PPT, Smartboard., SQL Plus			
		Practical 6: Implement SQL queries for Retrieving data using SELECT, WHERE, ORDER BY.	10 Min	SQL Plus			
		Practical Assignment-1 Problem Solving and Practice. (21-22 Questions)	45 Min	SQL Plus			
		Homework:					
		1. Create table EMPLOYEE, Add 25 Records and apply the below queries:					
		Create Table EMPLOYEE with fields: emp_id (INT, Primary Key), emp_name (VARCHAR(50)), department (VARCHAR(30)), salary (INT), and joining_date (DATE).					
		Q1. Insert a record into the EMPLOYEE table with the following details: emp_id = 201, emp_name = 'Rohan', department = 'HR', salary = 35000, joining_date = '2023-05-10'					
		Q2. Insert 4 more employee records into the EMPLOYEE table with different values for each column.					
6	1	Topic : DQL Commands, Clauses: Order By, Having, aand Aggregate Functions					
		Revision	15 Min	Discussion			
		DQL Command: SELECT with ORDER BY and HAVING Clauses	20 Min	PPT, Smartboard., SQL Plus			
		Practical 7: Implement SQL queries using aggregate functions with GROUP BY and HAVING	10 Min	SQL Plus			
		Practical Assignment-2 Problem Solving and Practice. (1-2 Questions)	40 Min	SQL Plus			
		Use of Data Dictionary in Oracle (e.g., USER_TABLES, ALL_TABLES).	15 Min	SQL Plus			
		Complete Pending Assignment and Pending Homework	20 Min	SQL Plus			



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7	1	Topics: Design Your College Database Title: Database Design Activity					
		- Students work in groups of 3–4. - Design a database for a college management system. - Identify entities, fields, records, and tables. - Create a data dictionary for all tables. - Present the database structure using charts or diagrams. Outcome: Understanding of data, metadata, database design, and data dictionaries.	115 Min	Database Design Activity			
		Activity Work Submission	05 Min	Submission			
8	1	Topics: DDL Commands, DML Commands, Data Dictionary					
		Revision	15 Min	Discussion			
		Practical 10: Implement SQL Queries to perform DDL Commands (Create 5 tables and perform all the DDL operations upon them).	20 Min	SQL Plus			
		Practical 11: Insert sample data into your tables using SQL INSERT statements. Verify the data has been inserted correctly.	20 Min	SQL Plus			
		Practical 16: Implement SQL queries for Understanding and browsing data dictionary: USER_TABLES, ALL_TABLES	15 Min	SQL Plus			
		Practical Assignment-2 Problem Solving and Practice. (3-4 Questions)	40 Min	SQL Plus			
		Homework:					
		1. Create a table named Student with appropriate columns and constraints such as Primary Key, NOT NULL, and UNIQUE. 2. Insert at least 5 records into the Student table using SQL INSERT statements. 3. Display all records from the Student table. 4. Retrieve the details of students belonging to a specific department. 5. Count the total number of students present in the Student table.					
9	1	Topics: DML Commands, DDL Commands, DQL Commands					
		Revision	15 Min	Discussion			
		Practical 12: Implement SQL Queries to perform DML Commands (Insert Minimum 10 records and perform modification operations upon them)	15 Min	SQL Plus			
		Practical 13: Practice DDL-create and DML-insert commands.	15 Min	SQL Plus			
		Practical 14: Retrieve data using SELECT command from tables.	15 Min	SQL Plus			
		Practical 15: Practice using SQL UPDATE and DELETE statements to modify data.	15 Min	SQL Plus			
		Practical Assignment-2 Problem Solving and Practice. (5-6 Questions)	30 Min	SQL Plus			
		Practical Questions Checking and Submission	15 Min	SQL Plus			
		Homework:					
1. Create a Student table with the fields: Student_ID, Student_Name, Department, Semester, and Marks. Insert at least 10 records into the table. 2. Retrieve the details of all students and display only Student_ID, Student_Name, and Marks. 3. Display the details of students who belong to the Computer Engineering department and have Marks greater than 70. 4. Update the Marks of the student with Student_ID = 105 by adding 5 bonus marks. Display the updated record. 5. Delete the record of the student with Student_ID = 110 and display the remaining records in the table. □							



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10	1	Topics: Open source DBMS V/s Commercial DBMS, DBMS Terminologies Title : Debate Activity and DBMS Vocabulary Crossword							
		Discussion groups conducted by selected student chairpersons and Topic given by faculty as Open source DBMS V/s Commercial DBMS	60 Min	Discussion					
		Students will solve a printed DBMS Vocabulary Crossword in class within 1 hour and submit the scanned copy through the ERP portal.	60 Min	DBMS Vocabulary Crossword Sheet					
Module-2 : Database System Architecture and Ensuring Data Integrity									
11	2	Topic : Architecture of Oracle, Components of DBMS and Data Integrity & Constraints							
		Architecture of Oracle: Instance and Database,	30 Min	PPT, Smartboard.					
		Components of DBMS (SGA, PGA, Background Processes, Data Files, Control Files)	30 Min	PPT, Smartboard.					
		Data Integrity and Constraints: Concept and Need of Constraints.	30 Min	PPT, Smartboard.					
		Practical 1: Apply primary keys and Not NULL constraints.	15 Min	SQL Plus					
		Practical 2: Implementing NOT NULL, CHECK, UNIQUE	15 Min	SQL Plus					
		Homework: 1. Create a table STUDENT(student_id INT PRIMARY KEY, student_name VARCHAR(50) NOT NULL, course VARCHAR(30), email VARCHAR(50)) and insert 5 records. 2. Create a table EMPLOYEE(emp_id INT PRIMARY KEY, emp_name VARCHAR(50) NOT NULL, department VARCHAR(30), salary INT) and test insertion of a record with NULL employee name. 3. Create a table PRODUCT(product_id INT PRIMARY KEY, product_name VARCHAR(50) NOT NULL, price DECIMAL(8,2) CHECK(price > 0), quantity INT CHECK(quantity >= 0)) and insert valid and invalid records. 4. Create a table CUSTOMER(customer_id INT PRIMARY KEY, customer_name VARCHAR(50) NOT NULL, email VARCHAR(50) UNIQUE, city VARCHAR(30)) and verify that duplicate email addresses are not allowed. 5. Create a table COURSE(course_id INT PRIMARY KEY, course_name VARCHAR(50) NOT NULL, duration_months INT CHECK(duration_months BETWEEN 1 AND 24), course_code VARCHAR(10) UNIQUE) and insert records to test all constraints.							
12	2	Topics: About DBA, DBA Role and Responsibilities, Constraints: Domain, Entity and Referential							
		Database administrator (DBA)	15 Min	PPT, Smartboard.					
		DBA Role and Responsibilities	15 Min	PPT, Smartboard.					
		Domain Constraints: NOT NULL, CHECK	15 Min	PPT, Smartboard.					
		Entity Constraints: UNIQUE, PRIMARY KEY	15 Min	PPT, Smartboard, SQL Plus					
		Referential Constraints: FOREIGN KEY	15 Min	PPT, Smartboard, SQL Plus					
		Practical 3: Using IN operator in queries	15 Min	SQL Plus					
		Practical 4: Perform queries involving predicates LIKE, BETWEEN, IN etc.	15 Min	SQL Plus					
		Practical 6: Add constraints like Primary key, NOT NULL and UNIQUE to your tables.	15 Min	SQL Plus					
		Homework: Table: STUDENT(Student_ID, Student_Name, Department, Marks, City) 1. Retrieve the details of students whose Department is either 'CSE' or 'IT' using the IN operator. 2. Display the records of students whose names start with 'A', Marks are between 60 and 90, and City is in ('Ahmedabad', 'Surat') using LIKE, BETWEEN, and IN predicates. 3. Modify the STUDENT table by adding PRIMARY KEY on Student_ID, NOT NULL on Student_Name, and UNIQUE on Email_ID (assume Email_ID column exists), then verify the applied constraints.							



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13	2	Topic: Database system architecture, Data abstraction, Data Independence, Operators: IN, LIKE, BETWEEN								
		Revision	05 Min	Discussion						
		Database system architecture	20 Min	PPT, Smartboard.						
		Data Abstraction	15 Min	PPT, Smartboard.						
		Data Independence (Logical & Physical)	15 Min	PPT, Smartboard.						
		IN Operator	10 Min	PPT, Smartboard, SQL Plus						
		LIKE	10 Min	PPT, Smartboard, SQL Plus						
		BETWEEN	10 Min	PPT, Smartboard, SQL Plus						
		Practical 7: Use foreign keys to establish relationships between tables.	15 Min	SQL Plus						
		Practical 8: Implement practical 1 again with referential constraint.	20 Min	SQL Plus						
		Homework: 1. Create table EMPLOYEE (emp_id INT PK, emp_name VARCHAR(50), department VARCHAR(30), salary INT, joining_date DATE) and insert at least 10 records. 2. Write a query to display employees whose department is 'HR', 'Sales', or 'IT' using the IN operator. 3. Write a query to display employees whose names start with the letter 'A' using the LIKE operator. 4. Write a query to display employees whose salary is between 30000 and 60000 using the BETWEEN operator. 5. Write a query to display employees whose department is 'IT' and salary is between 40000 and 70000 using a combination of IN and BETWEEN operators.								
14	2	TPA Full Session.	120 Min	N/A						
Topics: Flipped Classroom Activity										
15	2	Topic: Types of DBMS, Aggregate Functions, Inbuilt Functions								
		Revision	15 Min	Discussion						
		Types of DBMS – Centralized and Distributed	30 Min	PPT, Smartboard.						
		Aggregate Functions	20 Min	PPT, Smartboard, SQL Plus						
		Inbuilt Functions: Numeric	20 Min	PPT, Smartboard, SQL Plus						
		Practical 10: Implement SQL queries using Numeric functions like abs, ceil, power, mod, round, trunc, sqrt, etc.	20 Min	SQL Plus						
		Practical 13: Implement SQL queries using Group functions like Avg, Min, Max, Sum, Count, etc.	15 Min	SQL Plus						
		Homework: EMPLOYEE (emp_id INT PK, emp_name VARCHAR(50), department VARCHAR(30), salary INT, joining_date DATE) 1. Write SQL queries using ROUND() to display employee salaries rounded to the nearest thousand. 2. Write SQL queries using MOD() to find the remainder when employee salaries are divided by 1000. 3. Write SQL queries using POWER() and SQRT() functions on employee salary values. 4. Write SQL queries using ABS(), CEIL(), and TRUNC() functions and display the results. 5. Write SQL queries to find the COUNT() of employees in the table. 6. Write SQL queries to find the SUM() and AVG() of employee salaries. 7. Write SQL queries to find the MIN() and MAX() salary among employees. 8. Write SQL queries using GROUP BY department to display the department-wise COUNT(), SUM(), and AVG() salary.								
16	2	Topics: Oracle Architecture and DBMS Components Title : Build DBMS Model (Group Activity)								
		Divide students into groups of 5–6 and have each group create a 3D model, chart, or poster illustrating the components of an Oracle Database Architecture, including Oracle Instance, Database, SGA, PGA, Background Processes, Data Files, Control Files, and DBA.	90 Min	3D Model Creation						
		Students work in teams to explain the role of each Oracle Database Architecture component as system architects, fostering experiential learning, teamwork, critical thinking, and communication skills.	30 Min	Presentation						



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17	2	Topic: Inbuilt Functions					
		Revision	15 Min	Discussion			
		Inbuilt Functions: Date	35 Min	PPT, Smartboard, SQL Plus			
		Practical 9: Implement SQL queries using Date functions like add-months, months-between, round, next day, truncate, etc.	30 Min	SQL Plus			
		Practical Assignment-3 Problem Solving and Practice. (1-10 Questions)	50 Min	SQL Plus			
		Homework: EMPLOYEE (emp_id INT PK, emp_name VARCHAR(50), department VARCHAR(30), salary INT, joining_date DATE) 1. Write SQL queries using ADD_MONTHS() to display the joining date after adding 6 months to each employee's joining date. 2. Write SQL queries using MONTHS_BETWEEN() to calculate the number of months between the employee's joining date and the current date. 3. Write SQL queries using NEXT_DAY() to display the next Monday after each employee's joining date. 4. Write SQL queries using ROUND() and TRUNC() on employee joining dates to display rounded and truncated date values.					
18	2	Topic: Inbuilt Functions					
		Revision	15 Min	Discussion			
		Inbuilt Functions:String	30 Min	PPT, Smartboard, SQL Plus			
		Practical 11: Implement SQL queries and String Functions like initcap, lower, upper, rtrim, replace, substring, instr, etc.	35 Min	SQL Plus			
		Practical Assignment-3 Problem Solving and Practice. (11-25 Questions)	40 Min	SQL Plus			
		Homework: EMPLOYEE (emp_id INT PK, emp_name VARCHAR(50), department VARCHAR(30), salary INT, joining_date DATE) 1. Write SQL queries using UPPER() and LOWER() to display employee names in uppercase and lowercase. 2. Write SQL queries using INITCAP() to display employee names with the first letter of each word capitalized. 3. Write SQL queries using SUBSTR() and INSTR() to extract characters from employee names and find the position of a specific character. 4. Write SQL queries using RTRIM() and REPLACE() to remove trailing characters and replace specific text in employee names or department names.					
19	2	Topic: Conversion Functions					
		Revision	20 Min	Discussion			
		Practical 12: Implement SQL queries using Conversion functions like to-char, to-date, to-number.	35 Min	PPT, Smartboard, SQL Plus			
		Practical Assignment-3 Problem Solving and Practice. (26-50 Questions)	50 Min	SQL Plus			
		Explanation of SEE-2 and ALA of Module 2: Student Instructions for Performing the Activity and Submitting the Required Work	15 Min	Explanation			
		Homework: Table: EMPLOYEE(Emp_ID, Emp_Name, Salary, Join_Date, Bonus) 1. Display employee names and their joining dates in the format 'DD-MON-YYYY' using the TO_CHAR function. 2. Display employee names and salaries with a currency symbol and comma separators using the TO_CHAR function. 3. Convert the string '15-AUG-2024' into a date using the TO_DATE function and display the result. 4. Find all employees who joined after '01-JAN-2023' by converting the string date using the TO_DATE function. 5. Convert the string numbers '1500' and '2500' into numeric values using the TO_NUMBER function and display their sum.					



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20	2	Topic: Inbuilt functions: numeric, date, and string					
		Revision	20 Min	Discussion			
		Practical 5: Practice with numeric, date, and string inbuilt functions	35 Min	SQL Plus			
		Practical Assignment-4 Problem Solving and Practice. (1-25 Questions)	65 Min	SQL Plus			
		Homework: Table: EMPLOYEE(Emp_ID, Emp_Name, Salary, Join_Date, Department) 1. Display employee names in uppercase and calculate the length of each employee name. 2. Display employee names along with the first three characters of their names using a string function. 3. Display employee salaries after rounding them to the nearest integer using a numeric function. 4. Display the current system date and the date 30 days after today using date functions. 5. Calculate the number of months each employee has worked in the organization from their joining date until today.					
21	2	Topics: Constraints and Data Integrity Title : Database Integrity Detective Challenge					
		Provide a faulty tables containing NULL values, duplicate IDs, invalid marks, and missing foreign keys for data validation and error identification exercises.	10 Min	Instructor Briefing			
		Students identify violated constraints and propose corrective actions while applying NOT NULL, UNIQUE, PRIMARY KEY, FOREIGN KEY, and CHECK constraints.	70 Min	Indetifying Corrective Actions			
		Students develop problem-solving and analytical thinking skills by evaluating data inconsistencies, identifying constraint violations, and recommending appropriate solutions.	40 Min	Evaluation			
Sessional Exam - I (Module 1 & 2)							
Module-3 : Advanced SQL and Normalization							
22	3	Topics: Advanced Joins: Inner, Outer, Self Join					
		Advanced Joins: Inner, Outer, Self Join	20 Min	PPT, Smartboard.			
		Advanced Joins: Inner, Outer, Self Join Example	25 Min	PPT, Smartboard, SQL Plus			
		Displaying data from Multiple Tables (join)	25 Min	PPT, Smartboard.			
		Practical 1:To understand and implement various types of SQL joins — inner join, outer join (left, right, full), and self join	15 Min	PPT, Smartboard.			
		Practical 2:To learn how to retrieve and display combined data from multiple relational database tables using different types of SQL joins, thereby understanding data relationships and improving data querying efficiency.	15 Min	SQL Plus			
		Practical Assignment-5 Problem Solving and Practice Joins. (1-10 Questions)	20 Min	SQL Plus			
		Homework: DEPARTMENT(Dept_ID NUMBER(3) PK, Dept_Name VARCHAR2(30), Location VARCHAR2(20)) EMPLOYEE(Emp_ID NUMBER(5) PK, Emp_Name VARCHAR2(30), Salary NUMBER(8,2), Dept_ID NUMBER(3) FK REFERENCES DEPARTMENT(Dept_ID), Manager_ID NUMBER(5)) 1. Display employee names along with their department names using an INNER JOIN. 2. Display all departments and their employees, including departments with no employees using a LEFT OUTER JOIN. 3. Find departments that do not have any employees. 4. Display employee names along with their manager names using a SELF JOIN. 5. Find employees whose salary is greater than the salary of their manager.					



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23	3	Topics: Subqueries, Nested Queries, Correlated Subqueries					
		Subqueries, Nested Queries, Correlated Subqueries	20 Min	PPT, Smartboard.			
		Subqueries, Nested Queries, Correlated Subqueries Example	15 Min	PPT, Smartboard, SQL Plus			
		Retrieve data from multiple tables using sub queries (multiple, nested, correlated).	15 Min	SQL Plus			
		SQL Queries: EXISTS, ANY, ALL	10 Min	PPT, Smartboard, SQL Plus			
		Practical 3: To practice the use of different types of subqueries — multiple-row, nested, and correlated — in SQL, enabling complex data extraction from multiple related tables based on dynamic or dependent conditions.	20 Min	SQL Plus			
		Practical 4: To develop the skill to solve complex SQL queries using subqueries, including single-row and multiple-row subqueries, thereby enhancing query logic and data manipulation capabilities.	20 Min	SQL Plus			
		Practical 5: To explore the usage of conditional operators EXISTS, ANY, and ALL in SQL for querying datasets with conditions involving comparisons against subquery results, supporting advanced decision-making in data retrieval.	20 Min	SQL Plus			
		Homework: DEPARTMENT(Dept_ID NUMBER(3) PK, Dept_Name VARCHAR2(30), Location VARCHAR2(20)) EMPLOYEE(Emp_ID NUMBER(5) PK, Emp_Name VARCHAR2(30), Salary NUMBER(8,2), Dept_ID NUMBER(3) FK REFERENCES DEPARTMENT(Dept_ID), Manager_ID NUMBER(5)) 1. Display the names of employees who work in the department named 'IT' using a nested subquery. 2. Find employees whose salary is greater than the average salary of their department using a correlated subquery. 3. Display departments for which at least one employee exists using the EXISTS operator. 4. Find employees whose salary is greater than ANY salary of employees working in the HR department using the ANY operator. 5. Display employees whose salary is greater than ALL salaries of employees working in the Sales department using the ALL operator.					
24	3	Topics : Join and Subquery Title : . SQL Escape Room Game					
		Students solve SQL puzzles	10 Min	Instructor Briefing			
		• Level 1: INNER JOIN • Level 2: Subquery • Level 3: Correlated Subquery • Level 4: EXISTS Each correct query unlocks the next clue.	80 Min	Indetifying Corrective Actions			
		Students Feedback	30 Min	Experience Sharing			
25	3	Topics: Set Operations: UNION, INTERSECT, MINUS					
		Revision	15 Min	Discussion			
		Set Operations: UNION, INTERSECT, MINUS	35 Min	PPT, Smartboard.			
		Executing UNION, INTERSECT, MINUS	15 Min	SQL Plus			
		Practical 6: To understand and implement set operators — UNION, INTERSECT, and MINUS — for combining the results of multiple SQL queries and refining output by eliminating duplicates or performing comparisons between datasets.	20 Min	SQL Plus			
		Practical Assignment-5 Problem Solving and Practice Joins. (11-20 Questions)	35 Min	SQL Plus			
		Homework: CURRENT_EMPLOYEE(Emp_ID NUMBER(5) PK, Emp_Name VARCHAR2(30), Dept_ID NUMBER(3), Salary NUMBER(8,2)) FORMER_EMPLOYEE(Emp_ID NUMBER(5) PK, Emp_Name VARCHAR2(30), Dept_ID NUMBER(3), Salary NUMBER(8,2)) 1.. Display the names of all employees from both CURRENT_EMPLOYEE and FORMER_EMPLOYEE using UNION. 2. Display the names of employees who appear in both CURRENT_EMPLOYEE and FORMER_EMPLOYEE using INTERSECT. 3. Display the names of employees who are currently employed but are not present in FORMER_EMPLOYEE using MINUS. 4. Display the department IDs that exist in either CURRENT_EMPLOYEE or FORMER_EMPLOYEE using UNION. 5. Display the employee IDs that are present in FORMER_EMPLOYEE but not in CURRENT_EMPLOYEE using MINUS.					



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26	3	Topics: Views, Indexes, Sequences, Synonyms						
		Revision	15 Min	Discussion				
		Views, Indexes, Sequences, Synonyms	40 Min	PPT, Smartboard.				
		Creating and using Views, Indexes, Sequences	25 Min	SQL Plus				
		Practical 7:To demonstrate the creation and practical usage of database objects such as views (for virtual tables), indexes (for query performance optimization), and sequences (for auto-generating unique values), enhancing database design and efficiency.	40 Min	SQL Plus				
		Homework: EMPLOYEE(Emp_ID NUMBER(5) PRIMARY KEY, Emp_Name VARCHAR2(30), Salary NUMBER(8,2), Dept_ID NUMBER(3), Join_Date DATE) 1. Create a view named EMP_DETAILS that displays Emp_ID, Emp_Name, Salary, and Dept_ID for all employees. Retrieve data from the view. 2. Create an index on the Emp_Name column of the EMPLOYEE table and verify its creation. 3. Create a sequence named EMP_SEQ starting from 1001 and use it to insert a new employee record. Also create a synonym EMP for the EMPLOYEE table and display records using the synonym.						
27	3	Topics: Transaction Control Commands: COMMIT, ROLLBACK, SAVEPOINT						
		Revision	05 Min	Discussion				
		Transaction Control Commands: COMMIT, ROLLBACK, SAVEPOINT	30 Min	PPT, Smartboard.				
		Control Commands: COMMIT, ROLLBACK, SAVEPOINT Examples	25 Min	PPT, Smartboard.				
		Practical 8:Demonstrating Transactions: COMMIT, ROLLBACK, SAVEPOINT	20 Min	SQL Plus				
		Explanation of SEE-3 and ALA of Module 3: Student Instructions for Performing the Activity and Submitting the Required Work	10 Min	Explanation				
		Practical Assignment-5 Problem Solving and Practice Subqueries. (1-10 Questions)	30 Min	SQL Plus				
		Homework: EMPLOYEE(Emp_ID NUMBER(5) PRIMARY KEY, Emp_Name VARCHAR2(30), Salary NUMBER(8,2), Dept_ID NUMBER(3)) 1. Insert three employee records into the EMPLOYEE table, create a SAVEPOINT after the second insertion, insert one more record, and then ROLLBACK to the savepoint. 2. Update the salary of all employees in a specific department, COMMIT the changes, and verify that the updates are permanently saved. 3. Delete employees with a salary less than 20,000, create a SAVEPOINT, delete additional records, and then ROLLBACK to the savepoint to restore the deleted records after the savepoint.						
28	3	Topics: Introduction to Database Design, Functional Dependency, Normalization						
		Revision	15 Min	Discussion				
		Introduction to Database Design	15 Min	PPT, Smartboard.				
		Functional Dependency	15 Min	PPT, Smartboard.				
		Normalization: 1NF, 2NF, 3NF, BCNF	40 Min	PPT, Smartboard.				
		Comparison of 3NF and BCNF	20 Min	PPT, Smartboard.				
		Multi valued Dependency: 4NF & 5NF (Project Join NF),	15 Min	SQL Plus				
		Homework: STUDENT_COURSE(Student_ID NUMBER(5), Student_Name VARCHAR2(30), Course_ID NUMBER(5), Course_Name VARCHAR2(50), Faculty_ID NUMBER(5), Faculty_Name VARCHAR2(30), Skill VARCHAR2(30)) 1. The STUDENT_COURSE table contains repeating and multivalued data. Normalize the table up to 3NF and identify the functional dependencies involved. 2. Analyze the given table and determine whether it satisfies BCNF. If not, decompose it into BCNF-compliant relations and compare the resulting design with 3NF. 3. The table stores multiple skills for a student and multiple faculty assignments for a course. Identify the multivalued dependencies and decompose the table into 4NF and 5NF (Project Join Normal Form) relations.						



Gyanmanjari Innovative Unniversity

Gyanmanjari Degree Engineering College

Department of Computer Science and Engineering

Semester-3

Database Management System: Organizing, Querying, and Managing Data – BET1CE13305

Name of Faculty: Dr. Mansi C. Shanishwara

Hrs. Week: 6 hours

Micro Session Planner

Session no.	Module	Description	Time	Teacher aid	Planned date	Actual date	Remarks			
29	3	Topics:Transaction Processing, Database Backup & Recovery, ORACLE Utility – Import, Export.								
		Revision	10 Min	Discussion						
		Transaction Processing Introduction	20 Min	PPT, Smartboard.						
		ACID property	25 Min	Video, Explanation						
		Serializability of scheduling	10 Min	PPT, Smartboard.						
		Database Backup & Recovery	15 Min	PPT, Smartboard.						
		ORACLE Utility – Import, Export.	20 Min	PPT, Smartboard.						
		Practical 9:To perform database backup and recovery operations using export and import utilities, ensuring data protection and availability by understanding the process of safeguarding data in case of failure or loss.	20 Min	SQL Plus						
Homework: 1. Watch the video links of today's session available in Google Classroom.										
30	3	Topics:Practice Questions								
		Revision	15 Min	Discussion						
		Practical Assignment-6 Problem Solving and Practice. (1-27 Questions)	105 Min	SQL Plus						
		Homework: 1. Complete Pending Assignment Questions								
31	3	Topics: Normalization and Subqueries Title : Design, Normalize and Query								
		Form groups of 4 students and assign each group a real-world case study (Hospital Management System, Library Management System, or College ERP System) to analyze, design the database structure, identify functional dependencies, and normalize the data up to BCNF.	20 Min	Explanation, Definition Finalization						
		Design ER Diagram	30 Min	Lucidchart						
		Normalize up to BCNF	20 Min	Microsoft Word						
		Write 5 SQL Queries using Joins/Subqueries	20 Min	SQL Plus						
		Present findings	30 Min	Presentation						
Module-4 : PL/SQL and Database Security										
32	4	Topics: PL/SQL and Variables								
		Revision	10 Min	Discussion						
		Introduction to PL/SQL	15 Min	PPT, Smartboard.						
		PL/SQL Block Structure	20 Min	PPT, Smartboard, SQL Plus						
		Variables	15 Min	PPT, Smartboard.						
		Practical 1:Implement basic PL/SQL blocks.	30 Min	SQL Plus						
		Practical Assignment-7 Problem Solving and Practice. (1-5 Questions)	30 Min	SQL Plus						
		Homework: 1. Create Table EMPLOYEE (emp_id NUMBER(5) PRIMARY KEY, emp_name VARCHAR2(50), department VARCHAR2(30), salary NUMBER(8), joining_date DATE); 2. Write a PL/SQL block to display "Welcome to PL/SQL" using DBMS_OUTPUT.PUT_LINE. 3. Write a PL/SQL block to declare a variable v_name, assign your name, and display it. 4. Write a PL/SQL block to declare two numbers, add them, and display the result. 5. Write a PL/SQL block to fetch and display the emp_name of an employee whose emp_id = 101 from the EMPLOYEE table. 6. Write a PL/SQL block to count the total number of employees in the EMPLOYEE table and display the result.								



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Micro Session Planner

Session no.	Module	Description	Time	Teacher aid	Planned date	Actual date	Remarks
33	4	Topics: Advantages of PL/SQL and IF, LOOP, CASE					
		Revision	10 Min	Discussion			
		Control Structures Syntax	10 Min	PPT, Smartboard.			
		Advantages of PL/SQL	10 Min	PPT, Smartboard.			
		IF STATEMENT	15 Min	PPT, Smartboard.			
		WHILE LOOP	15 Min	PPT, Smartboard.			
		CASE	15 Min	PPT, Smartboard.			
		Practical 2: Implement PL/SQL programs using IF STATEMENT.	15 Min	SQL Plus			
		Practical 3: Implement PL/SQL programs using WHILE LOOP.	15 Min	SQL Plus			
		Practical 4: Implement PL/SQL programs using IF, CASE, and loops.	15 Min	SQL Plus			
		Homework:					
		1. Create Table EMPLOYEE (emp_id NUMBER(5) PRIMARY KEY, emp_name VARCHAR2(50), department VARCHAR2(30), salary NUMBER(8), joining_date DATE);					
		2. Write a PL/SQL block to display an employee's salary category (High, Medium, or Low) using a CASE statement and display the result only if the employee exists using an IF statement.					
34	4	Topics: Cursors and its Types					
		Revision	10 Min	Discussion			
		Cursors: Implicit	15 Min	PPT, Smartboard.			
		Cursors: Explicit	15 Min	PPT, Smartboard.			
		Cursors: Cursor FOR Loop	15 Min	PPT, Smartboard.			
		Cursors: Parameterized Cursor	15 Min	PPT, Smartboard.			
		Practical 5: Implement PL/SQL programs with Cursor implementation (all types).	30 Min	SQL Plus			
		Practical Assignment-7 Problem Solving and Practice. (6-10 Questions)	20 Min	SQL Plus			
		Homework:					
		1. Create Table EMPLOYEE (emp_id NUMBER(5) PRIMARY KEY, emp_name VARCHAR2(50), department VARCHAR2(30), salary NUMBER(8), joining_date DATE);					
		2. Write a PL/SQL block using an Implicit Cursor to update the salary of an employee and display the number of rows affected.					
		3. Write a PL/SQL block using an Explicit Cursor to display the employee ID and employee name of all employees.					
		4. Write a PL/SQL block using a Cursor FOR Loop to display the employee name and department of all employees.					
		5. Write a PL/SQL block using a Parameterized Cursor to display employee details for a given department.					
		6. Write a PL/SQL block using a Cursor FOR Loop to display employees whose salary is greater than 50,000.					



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Hrs. Week: 6 hours

Micro Session Planner

Session no.	Module	Description	Time	Teacher aid	Planned date	Actual date	Remarks
35	4	Topics: IF, WHILE, CASE, CURSOR					
		Revision PL/SQL Block	15 Min	Discussion			
		Revision IF	15 Min	PPT, Smartboard.			
		Revision While	15 Min	PPT, Smartboard.			
		Revision Case	15 Min	PPT, Smartboard.			
		Revision of Cursor	15 Min	PPT, Smartboard.			
		Practical Assignment-7 Problem Solving and Practice. (11-15 Questions)	30 Min	SQL Plus			
		Explanation of SEE-4 and CCE of Module 4: Student Instructions for Performing the Activity and Submitting the Required Work	15 Min	Explanation			
		Homework: 1. Create Table STUDENT (student_id NUMBER(5) PRIMARY KEY, student_name VARCHAR2(50), course VARCHAR2(30), marks NUMBER(3), admission_date DATE); 2. Write a PL/SQL block to fetch and display the student name and marks for a given student ID. 3. Write a PL/SQL block to check whether a student's marks are greater than 70 and display an appropriate message. 4. Write a PL/SQL block using a WHILE LOOP to display student IDs from 1 to 5. 5. Write a PL/SQL block to classify a student's result as Distinction, First Class, Pass, or Fail based on marks. 6. Write a PL/SQL block to display the student ID, student name, and course of all students using a cursor.					
36	4	Topics: PL/SQL Block and Exception Handling Title: PL/SQL Debugging and Exception Handling Workshop					
		Form groups of 4 students. Provide PL/SQL programs containing logical and runtime errors. Identify errors and classify them.	10 Min	Instructor Briefing			
		Implement predefined exception handlers. Create user-defined exceptions for business rules.	80 Min	Activity Execution			
		Present the corrected solutions.	30 Min	Presentation			
37	4	Topics: Exception Handling, SQL v/s PL/SQL					
		Revision	15 Min	Discussion			
		Exception Handling Introduction	15 Min	PPT, Smartboard.			
		Predefined Exceptions	15 Min	PPT, Smartboard.			
		User-defined Exceptions	15 Min	PPT, Smartboard.			
		Handling Raised Exceptions	15 Min	PPT, Smartboard.			
		SQL vs PL/SQL	15 Min	PPT, Smartboard.			
		Practical 6: Implement PL/SQL programs using Exception handling examples.	30 Min	SQL Plus			
		Homework: 1. Create Table EMPLOYEE (emp_id NUMBER(5) PRIMARY KEY, emp_name VARCHAR2(50), department VARCHAR2(30), salary NUMBER(8), joining_date DATE); 2. Write a PL/SQL block to display the details of an employee for a given emp_id and handle the exception if the employee does not exist. 3. Write a PL/SQL block to find the salary of an employee and handle the predefined exception when no record is found. 4. Write a PL/SQL block to raise a user-defined exception if an employee's salary is less than 10,000. 5. Write a PL/SQL block to check whether an employee exists in the EMPLOYEE table and display an appropriate exception message if not found. 6. Write a PL/SQL block to raise and handle a custom exception when an employee's salary exceeds 1,00,000.					



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Micro Session Planner

Session no.	Module	Description	Time	Teacher aid	Planned date	Actual date	Remarks
38	4	Topics: Database Security, DAC, MAC					
		Revision	10 Min	Discussion			
		Database Security: Authentication, Authorization and Access control	30 Min	PPT, Smartboard.			
		DAC (Discretionary Access Control)	25 Min	PPT, Smartboard.			
		MAC (Mandatory Access Control)	25 Min	PPT, Smartboard.			
		Practical Assignment-7 Problem Solving and Practice. (16-20 Questions)	30 Min	SQL Plus			
		Homework: Watch Videos of todays Session Database Security: Authentication, Authorization and Access Control => https://www.youtube.com/watch?v=4M2M9-4QdD8 Discretionary Access Control (DAC) => https://www.youtube.com/watch?v=9Qw5Y8jQfYQ Mandatory Access Control (MAC) => https://www.youtube.com/watch?v=K717JQJ2Y7A DAC vs MAC vs RBAC (Very Useful) => https://www.youtube.com/watch?v=V4x7Lx6fM4Q Complete DBMS Security Playlist (Gate Smashers) => https://www.youtube.com/playlist?list=PLxCzCOWd7aiGfD0G3f4iA5q7J7vK6q1YI DBMS Security Concepts (Neso Academy) => https://www.youtube.com/results?search_query=neso+academy+database+security					
39 (AI)	4	Topics: Authentication, Authorization, DAC, MAC Title: AI-Powered Database Security Audit					
		Provide a Hospital or College ERP database scenario.	10 Min	Instructor Briefing, ChatGPT, AI Assistants			
		1. Use AI tools to suggest security policies and user roles. 2. Verify AI recommendations against database security principles. 3. Design an access-control matrix.	80 Min	Group Formation, Execution			
		Present security risks and improvements. Student Outcomes: Cybersecurity Awareness, Research Skills, Collaborative Learning	30 Min	AI-assisted Research, Presentation			
40	4	Topics: SQL Injection					
		Revision	10 Min	Discussion			
		SQL injection.	30 Min	PPT, Smartboard.			
		SQL injection Examples	20 Min	SQL Plus			
		Practical Assignment-7 Problem Solving and Practice. (21-25 Questions)	40 Min	SQL Plus			
		Doubt Solving of Concept, Assignment and Practical Questions	20 Min	SQL Plus			
		Homework: 1. Write a query to demonstrate how an SQL Injection attack can bypass a login form that uses username and password without input validation. 2. Identify the SQL Injection vulnerability in the following query and suggest a secure alternative: SELECT * FROM EMPLOYEE WHERE emp_name = 'user_input';					



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Micro Session Planner

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41	4	Topics: PL/SQL					
		PL/SQL programs practice	60 Min	SQL Plus			
		Practical Assignment-7 Problem Solving and Practice. (26-30 Questions)	40 Min	SQL Plus			
		Doubt Solving of Concept, Assignment and Practical Questions	20 Min	Discussion			
		Homework: 1. Create Table PRODUCT (product_id NUMBER(5) PRIMARY KEY, product_name VARCHAR2(50), category VARCHAR2(30), price NUMBER(8), stock NUMBER(5)); 2. Write a PL/SQL block to display the product name and price for a given product ID. 3. Write a PL/SQL block to increase the price of all products in a given category by 10% and display the number of rows updated. 4. Write a PL/SQL block to display the product ID, product name, and category of all products using a cursor.					
42 (AI)	4	Topics: Complete Module Title: AI-Generated Mini Project Development					
		Assign a domain (Hospital, Library, Banking, College ERP).	15 Min	Instructor Briefing			
		Use AI tools to generate: o Database schema o PL/SQL procedures o Security policies o Exception handling mechanisms	55 Min	Activity Execution			
		Validate all AI-generated content.	20 Min	Student Skill Enhancement			
		Each group Demonstrate the working solution and Reflect on the strengths and limitations of AI assistance. Student Outcome: Project-Based Learning, Industry Readiness, AI Literacy, Problem Solving	30 Min	Presentation			
Module-5: Advanced Concepts of SQL & Access Control							
43	5	Topics: Stored Procedures and Stored Functions					
		Revision	10 Min	PPT, Smartboard.			
		Stored Procedures	20 Min	PPT, Smartboard.			
		Stored Functions	20 Min	PPT, Smartboard.			
		Stored Procedures Examples	20 Min	SQL Plus			
		Stored Functions Examples	20 Min	SQL Plus			
		Practical 1:Creating stored procedures and functions	30 Min	SQL Plus			
		Homework: 1. Create Table EMPLOYEE (emp_id NUMBER(5) PRIMARY KEY, emp_name VARCHAR2(50), department VARCHAR2(30), salary NUMBER(8), joining_date DATE); 2. Write a stored procedure to display the employee name and salary for a given employee ID. 3. Write a stored function to return the total salary of all employees from the EMPLOYEE table.					
44	5	Topics: Stored Procedures and Stored Functions					
		Revision	10 Min	Discussion			
		Practical Assignment-8 Problem Solving and Practice. (1-20 Questions)	110 Min	SQL Plus			
		Homework: 1. Create Table STUDENT (student_id NUMBER(5) PRIMARY KEY, student_name VARCHAR2(50), course VARCHAR2(30), marks NUMBER(3), admission_date DATE); 2. Write a stored procedure to display the student name and marks for a given student ID. 3. Write a stored function to return the average marks of all students from the STUDENT table.					



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Micro Session Planner

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45	5	Topics: Packages					
		Revision	10 Min	Discussion			
		Packages	20 Min	PPT, Smartboard.			
		Packages Example	20 Min	SQL Plus			
		Practical 2: Creating and using packages	30 Min	SQL Plus			
		Practical Assignment-8 Problem Solving and Practice. (21-22 Questions)	40 Min	SQL Plus			
		Homework: 1. Create Table PRODUCT (product_id NUMBER(5) PRIMARY KEY, product_name VARCHAR2(50), category VARCHAR2(30), price NUMBER(8), stock NUMBER(5)); 2. Create a package that contains a procedure to display the details of a product for a given product ID. 3. Create a package that contains a function to return the total stock available for all products. 4. Create a package with a procedure to update the price of a product and display a success message. 5. Create a package with a function to return the highest product price from the PRODUCT table. 6. Create a package containing both a procedure to display product details and a function to return the average product price.					
46	5	Topics: Cursor					
		Revision	20 Min	Discussion			
		Practical 3: Implement Cursor	40 Min	SQL Plus			
		Practical Assignment-8 Problem Solving and Practice. (23-24 Questions)	20 Min	SQL Plus			
		Doubt Solving of Concept, Assignment and Practical Questions	40 Min	Discussion			
		Homework: 1. Create Table EMPLOYEE (emp_id NUMBER(5) PRIMARY KEY, emp_name VARCHAR2(50), department VARCHAR2(30), salary NUMBER(8), joining_date DATE); 2. Write a PL/SQL program using an Explicit Cursor to display the employee ID, employee name, and salary of all employees. 3. Write a PL/SQL program using a Cursor FOR LOOP to display the details of employees belonging to the IT department. 4. Write a PL/SQL program using a Parameterized Cursor to display employee details for a given department.					
47	5	Topics: Triggers and types of Triggers, Users – Create & Delete User					
		Revision	10 Min	Discussion			
		Triggers Introduction	10 Min	PPT, Smartboard.			
		Types of Triggers	20 Min	PPT, Smartboard.			
		Users – Create User	10 Min	PPT, Smartboard.			
		Users – Delete User	10 Min	PPT, Smartboard.			
		Testing triggers: BEFORE, AFTER, INSTEAD OF (Row/Statement-level)	20 Min	PPT, Smartboard.			
		Practical 4: Implement Trigger	20 Min	SQL Plus			
		Practical Assignment-8 Problem Solving and Practice. (25-26 Questions)	20 Min	SQL Plus			
		Homework: 1. Create Table EMPLOYEE (emp_id NUMBER(5) PRIMARY KEY, emp_name VARCHAR2(50), department VARCHAR2(30), salary NUMBER(8), joining_date DATE); 2. Create a BEFORE INSERT trigger to prevent inserting an employee record with a salary less than 10,000. 3. Create an AFTER UPDATE trigger to display a message whenever an employee's salary is updated. 4. Create a BEFORE DELETE trigger to prevent deletion of employee records from the EMPLOYEE table.					



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48	5	Topics: Stored Procedures and Stored Functions, Packages, Trigger					
		Revision Stored Procedures and Stored Functions	20 Min	Discussion			
		Revision Packages	10 Min	PPT, White Board			
		Revision Trigger	10 Min	PPT, White Board			
		Practical 5:Writing and testing triggers: BEFORE, AFTER, INSTEAD OF (Row/Statement-level)	30 Min	SQL Plus			
		Practical 6:Creating and managing users: CREATE and DELETE USER	30 Min	SQL Plus			
		Explanation of SEE-5 and ALA of Module 5: Student Instructions for Performing the Activity and Submitting the Required Work	20 Min	Explanation			
		Homework: Complete the ALA-5					
49	5	Topic: PL/SQL Title: SQL Escape Room					
		Faculty given SQL puzzles	15 Min	Instructor Briefing			
		•Students must solve SQL puzzles within a time limit. •Each query result unlocks the next "room." •Final challenge requires combining multiple SQL concepts.	90 Min	Solve puzzles			
		Students Feedback	15 Min	Discussion			
50	5	Topics: Data Control Language (DCL) Commands: Grant and Revoke, Transaction Processing Language (TPL): BEGIN TRANSACTION, COMMIT, ROLLBACK, SAVEPOINT, ROLLBACK TO SAVEPOINT					
		Data Control Language (DCL) Commands	10 Min	Discussion			
		Grant and Revoke	20 Min	PPT, Smartboard.			
		Transaction Processing Language (TPL)	10 Min	PPT, Smartboard.			
		BEGIN TRANSACTION	10 Min	PPT, Smartboard.			
		COMMIT, ROLLBACK and SAVEPOINT	30 Min	PPT, Smartboard, SQL Plus			
		ROLLBACK TO SAVEPOINT	10 Min	PPT, Smartboard.			
		Practical 7: Using GRANT and REVOKE	30 Min	SQL Plus			
		Homework: 1. Create Table ACCOUNT (account_id NUMBER(5) PRIMARY KEY, customer_name VARCHAR2(50), balance NUMBER(10), account_type VARCHAR2(20), opening_date DATE); 2. Insert three records into the ACCOUNT table, create a SAVEPOINT after the second insert, insert one more record, and then roll back to the SAVEPOINT. 3. Update the balance of multiple accounts, use COMMIT to save the changes permanently, and verify the updated records. 4. Delete a record from the ACCOUNT table, use ROLLBACK before COMMIT, and verify that the deleted record is restored.					



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Micro Session Planner

Session no.	Module	Description	Time	Teacher aid	Planned date	Actual date	Remarks
51	5	Topics: Locking Strategies: Implicit Locking and Explicit Locking, LOCK TABLE Command					
		Revision	10 Min	Discussion			
		Locking Strategies Introduction	10 Min	PPT, Smartboard.			
		Implicit Locking	20 Min	PPT, Smartboard.			
		Explicit Locking	20 Min	PPT, Smartboard.			
		LOCK TABLE Command	15 Min	SQL Plus			
		Practical 8:Demonstrating transaction control with SAVEPOINT and rollback	25 Min	SQL Plus			
		Practical Assignment-8 Problem Solving and Practice. (27-28 Questions)	20 Min	SQL Plus			
		Homework: Complete your pending work					
52	5	Topics: ORACLE Database Privileges & Roles: System & Object Privileges, Assigning, Viewing, Revoking, Roles – Create, Grant, View & Delete					
		ORACLE Database Privileges & Roles	10 Min	Discussion			
		System Privileges	10 Min	PPT, Smartboard.			
		Object Privileges	10 Min	PPT, Smartboard.			
		Assigning Privileges	10 Min	PPT, Smartboard.			
		Viewing Privileges	20 Min	PPT, Smartboard.			
		Revoking Privileges	20 Min	PPT, Smartboard.			
		Roles – Create, Grant, View & Delete	20 Min	PPT, Smartboard.			
		Practical Assignment-8 Problem Solving and Practice. (29 number Questions)	20 Min	SQL Plus			
		Homework: 1. Create a new user, grant the CREATE SESSION and CREATE TABLE system privileges, and verify the assigned privileges. 2. Grant SELECT and UPDATE privileges on the EMPLOYEE table to another user, verify the privileges, and then revoke the UPDATE privilege. 3. Create a role named EMP_ROLE, grant SELECT and INSERT privileges on the EMPLOYEE table to the role, assign the role to a user, and then remove the role from that user.					
53 (AI)	5	Topics: Complete Module Title: Database Innovation Hackathon using AI					
		Develop a mini database solution collaboratively.	15 Min	Instructor Briefing			
		Teams design and implement a complete database application.	60 Min	Solve puzzles			
		Include procedures, functions, triggers, security, and reports.	15 Min	Discussion			
		Present the final solution.	30 Min	Presentation			
		Sessional Exam - II (Module 3 ,4 & 5)					

Signature of Subject Faculty

Signature of
Head of Department